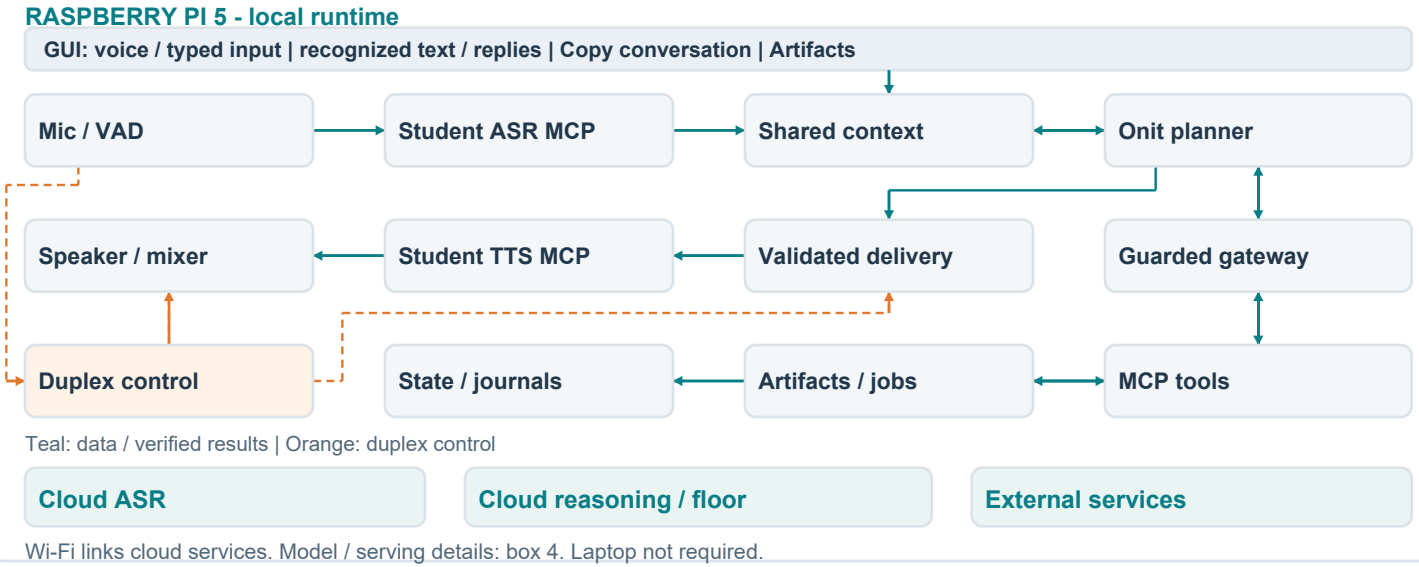
ME3 | Full-duplex voice agent

System architecture, versioned prompt and context engineering

1 System diagram: voice, reasoning and task control



Wi-Fi links cloud services. Model / serving details: box 4. Laptop not required.

2 Actual versioned prompt

"You are ME3, a general-purpose conversational assistant using Onit as the sole planner."

Voice and text share context. Onit selects tools; the host owns authorization, execution, cancellation and independent verification.

3 Context and task continuity

Voice and text are valid inputs. If speech is not recognized, type the request or correction.

Recognized text and replies are displayed; Copy conversation copies that exchange. Shared task context and action confirmations are preserved.

4 MCP services and model serving: local server does not mean local inference

Component	Pi implementation	Model / execution location
ASR	Student MCP, stdio	Groq whisper-large-v3-turbo / cloud; English-configured input
Reasoning / floor	Onit; tool-free advisory floor	gpt-6.1-sol / cloud; qwen/qwen3.8-27b / cloud
TTS / audio	Student MCP; Pipecat 1.12.0	Local Piper 1.8.0: English + female Spanish; VAD / SmartTurn on Pi
Tasks / UI	Guarded Pi-hosted MCP + basic GUI	Artifacts, search, calendars, media, Gmail, WhatsApp: scoped evidence

ME3 | Full-duplex voice agent

Measured evaluation, Pi-only demonstration and honest limitations

5 Evaluation: frozen candidate, updated course method

30 stage records (004); 14 disjoint behavior records (006 + 007). All declared selections recorded, not all passed. Original failed attempts retained; F7 PARTIAL. Frozen ME3 unchanged.

Metric	Eligible n	Mean	p95
Pi digital PCM handoff	21	6.439 s	7.357 s
Rig recorded reply onset	18	7.286 s	8.014 s
Endpointing	30	0.667 s	0.691 s
ASR finalization	30	0.790 s	1.315 s
ASR final to first text	24	4.773 s	6.169 s
TTS request to MCP PCM	21	0.507 s	0.694 s

ASR corpus WER: 17.68% (35 edits / 198 reference words, n=30). Means and p95 above use observed values only; see report for distributions and uncertainty.

Boundaries: Pi monotonic stage intervals. Digital PCM handoff and MCP PCM return are software observations, not physical audible onset/cessation. No physical microphone or speaker is used by the digital benchmark.

Stage no-reply labels: 12/30. Behavior: false turns 1/2; turn-taking miss 1/1; pause turns 0/1 each. Late/missing initial replies prevented overlaps: barge-in N=0. Judge semantics NOT_SCORED; physical audible stop and detection precision/recall NOT_MEASURED.

6 Inputs and configured task servers

Voice and text are valid inputs

Type unclear speech or exact details; Enter/Send. Both inputs share context; Copy conversation copies displayed text.

Spanish drafting/translation and female Spanish speech demonstrated; recognition remains English-configured.

Configured MCP task servers

1. Utilities
2. Shopping
3. Library
4. Music
5. Reminders
6. Research
7. YouTube
8. SoundCloud
9. WhatsApp
10. Workspace
11. Calling
12. Productivity
13. Gmail
14. Google Calendar

Configured scope is not blanket availability or live qualification. Account actions require exact review and separate confirmation.

Professor-directed demo on the Pi with Wi-Fi. Displayed text is not proof that all audio played.

R Challenges, proof and references

Natural pauses and barge-in: scoped owner demonstrations. Slow tool calls: contextual progress cues, not a speed claim. Messaging: verified final send/auto-close; redundant-close shutdown retained. Source restore: 7 focused checks, not a fresh-OS install.

Evidence: product/history reports; deployment provenance; Benchmarks/BENCHMARK_RESULTS.md. Course 57eaddbc188c; source 03d7a144917e. github.com/loreenannedeguzman/ME3_FULL_DUPLEX_VOICE_AGENT